Paleomagnetic Study of the Sudbury Basin[1]

PETER J. HOOD

Geophysics Laboratory, University of Toronto, Canada

Abstract. A reconnaissance paleomagnetic study has been carried out on the Precambrian nickel irruptive of the Sudbury basin. A total of 60 oriented samples was collected along a single traverse across the northern range and along four traverses across the southern range of the basin. The remanent magnetism (RM) vectors of the 266 specimen cubes cut from the samples were subsequently measured with a 'spinner' magnetometer. The RM directions for the south range are closely grouped on the Schmidt equal-area plot, and this fact indicates that there has been negligible relative rotary movement along the 16 miles of the south range sampled since the irruptive last cooled through the Curie point of its magnetic constituents. It was also found that the RM directions for the N and S ranges differ by approximately 40°, and this may be explained by assuming that there has been 40° of relative tilting about an E-W axis through the middle of the basin. In three of the S range traverses the susceptibility (κ) of the collected samples was higher on the S end of the traverse than on the N end, and in general the κ's obtained for the N range were much higher than those obtained on the S range. It would appear that the magnetic anomalies observed over the norite sections of the N and S ranges are mainly due to the presence of RM and to a lesser extent to an induced magnetization. The pole position calculated from the RM directions is dependent on the amount of assumed relative tilting. However if the rotation is equally divided between the two ranges, the resultant pole position is 38.4°N, 99.4°W, which is in the central U. S.

Introduction—general geology. The Sudbury area is one part of a geologic province extending more than 600 miles from the west end of Lake Superior, easterly through Sudbury and Cobalt, and on into Quebec. This vast belt is characterized by the occurrence of a quartz-diabase type of rock with which are associated important metallic ores.

Structural studies have shown that the nickel irruptive of the Sudbury basin, with which the nickel-copper deposits are associated, has the shape of an asymmetric basin (Fig. 1), its rim being exposed at the surface in the form of an elliptical ring elongated in a northeastern direction. The long and short axes of this ring are approximately 37 and 17 miles long, respectively.

The results from a gravity survey of the Sudbury basin [*Miller and Innes*, 1955] have appeared to support the contention by *Collins and Kindle* [1935] that the structure is approximately an asymmetric syncline.

Age determinations by the potassium-argon and rubidium-strontium methods [*Wetherill, Davis, and Aldrich*, 1957] have indicated that the nickel irruptive was emplaced 1200 to 1800 million years ago.

[1] Based on a thesis submitted in partial fulfillment of the requirements for the degree of Doctor of Philosophy at the University of Toronto.

The apparent thickness of the nickel irruptive, estimated from outcrops and from near-surface dips, ranges from 1 to 2 miles. It consists of an inner layer of micropegmatite separated by a transition zone from an outer, somewhat narrower layer of norite. Along most of the northern side of the basin the base of the norite near the surface dips inward at approximately 35° [*Thomson*, 1956], while at several places along the eastern and southeastern sides of the nickel irruptive the outer unfaulted contact is extremely steep, dipping outward in several places [*Lochhead*, 1955].

Collins [1934] described the norite as being a massive, dark grey rock, as coarse as ordinary granite, consisting of orthorhombic and monoclinic pyroxene, hornblende, biotite, plagioclase (basic labradorite to andesine), quartz, titaniferous magnetite, yellow sulfide (pyrite and pyrrhotite according to *Phemister* [1925]), titanite, and apatite. Collins found the micropegmatite to be more uniform that the norite. It is usually a dull, salmon-pink rock, somewhat coarser than the norite, composed chiefly of sodic plagioclase, orthoclase or microcline, quartz, and biotite.

The space inside the basin structure is occupied by a series of volcanic and sedimentary rocks, the layering of which outcrops in concentric

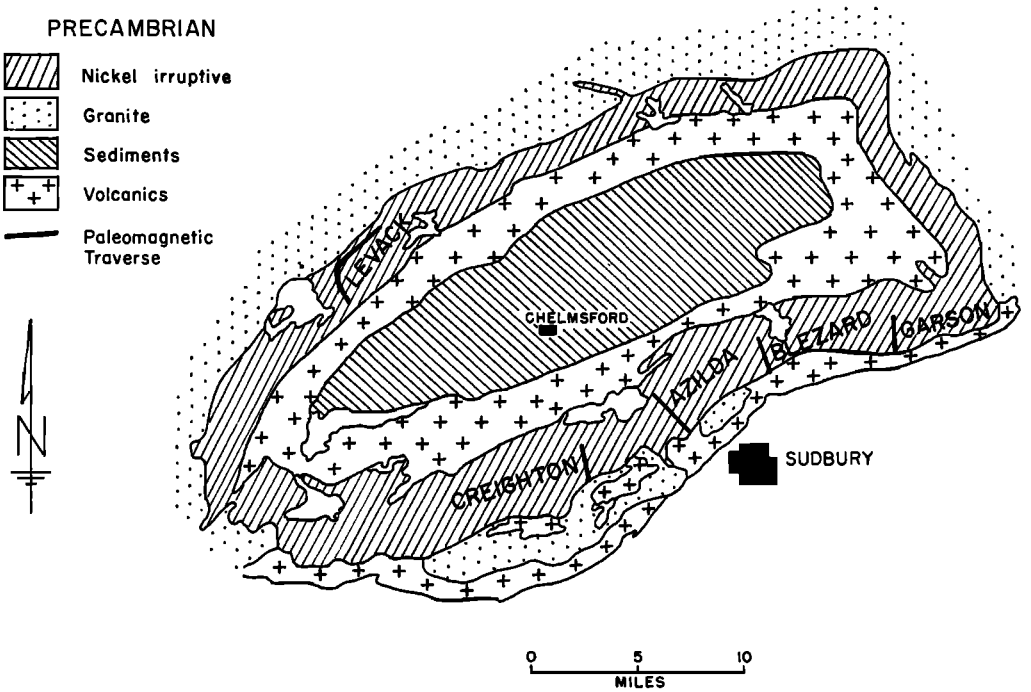


Fig. 1. Geologic map of the Sudbury basin showing sample traverses (After map 1956-1, Ontario Department of Mines).

rings. These volcanic and sedimentary rocks are considered to be conformable with the nickel irruptive.

North of the intrusive is an extensive area of granite, granite gneiss, and highly granitized sediments. To the south is a succession of volcanic, metamorphic, and sedimentary rocks, highly folded and faulted, striking in general east-northeast and dipping at steep angles mainly to the north.

Although the Sudbury area has been the subject of long and intensive studies by both mining companies and government organizations, the geology of the basin is still one of vigorous controversy. This study was carried out in the hope of throwing more light on the tectonic history of the area.

Sampling and measurement. A total of 60 oriented samples was collected for paleomagnetic studies from the nickel irruptive of the Sudbury basin during 1957; 37 of the samples were from the norite part of irruptive (Fig. 1). All the samples were taken either from road-cuts or from outcrops close to the highway. A suite of 18 samples was collected on the north range near the village of Levack, and the rest of the samples were collected on the south range at Creighton, Azilda, Blezard, and Garson.

In general, a sample sufficiently large to yield about five 1-inch cubes was collected, and the samples were oriented with a Brunton compass, care being taken to see that it was not appreciably deflected by the rock in the outcrop.

A total of 266 specimen cubes was cut with diamond saws from 59 of the samples. The natural remanent magnetization of each specimen cube was then determined in a 'spinner' magnetometer [*Hood*, 1958]. This apparatus was of the conventional 'rock generator' type in which the rock specimen is rotated opposite a fixed coil system at 45 revolutions per second, compressed air being used as a source of power.

Results. The directions of remanent magnetization of the specimens obtained from the five traverses have been plotted on the Schmidt equal-area projection [*Graham*, 1949] in Figure 2.

From the Azilda projection, it can be seen that the directions for the majority of the cubes fall in a fairly close group; however 11 specimens, all from the same road cut, do not fall in this group, and it is reasonable to suppose that there may have been some relative movement between

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

it and the main mass of norite. The results from this outcrop, therefore, have not been used in the calculation of the mean direction. Five of the cubes were demagnetized in an a-c field of about 125 oersteds using *Thellier and Rimbert's* [1954] method. The specimens, with one exception, were only slightly affected, which indicated that they possessed good stability and that the magnetization was of the thermoremanent type. The exception, which initially lay a little outside the group, had its direction considerably improved by a-c demagnetization, and was brought into the group. Four of the specimens from the anomalous outcrop previously mentioned, were also little affected by a-c demagnetization.

The Schmidt projections for the four traverses, Creighton, Blezard, Garson, and Levack, show fairly good groupings of the results. A number of the specimens from these traverses were also demagnetized in order to improve the grouping.

To determine the variation in susceptibility (κ) across the nickel irruptive, 'chip samples' were obtained from a number of the samples collected. The chips from a given sample were then crushed and divided into portions. The κ of each portion was measured with a calibrated susceptibility meter of the alternating current type, and the results were averaged to obtain the mean κ of the collected sample.

Figure 3 shows the results of the susceptibility measurements together with the average values of the intensity of remanent magnetization (RM) of the samples. Q is the Koenigsberger ratio, the ratio of remanent to induced magnetization; the value of the earth's field at Sudbury was taken to be 0.59 oersteds. Results from the Garson traverse were not included in Figure 3 because of the paucity of samples collected.

For the Azilda, Blezard, and Creighton traverses, it can be seen that all three physical properties are rather variable within the exposed section of the norite. In general, all these quantities decrease rapidly as the transition zone is approached. For all three traverses, κ tends to decrease from south to north across the norite, and for the Creighton and Blezard samples there appear to be well-defined maxima in Q and RM toward the center of the norite. Q remains well above the value of unity along all of the traverses and reaches values of between 10 and 20 within the norite.

Discussion of results and conclusions. The mean value of κ of the transition zone at Levack is about a factor of 100 higher than was obtained for the transition zone on the south range. However, the κ of the samples from the norite is similar to that obtained on the south range. Examination of an aeromagnetic map covering the area south from Levack to the south range [*Zietz and Henderson*, 1955] also shows that the magnetic anomalies produced by the transition zone on the north range have a much greater amplitude than those associated with the transition zone on the south range.

In addition, Q is less than unity for the transition zone near Levack. It would appear therefore, that the transition zone at Levack contains a greater percentage of magnetite than the transition zone on the south range, since κ is usually directly dependent on the magnetite content [*Mooney and Bleifuss*, 1953; *Runcorn*, 1956]. Thus the transition zone at Levack also contains a rather higher percentage of magnetite than the norite and micropegmatite from which it is thought to be formed. This is consistent with the results of *Collins* [1934], who showed that at the junction of the norite and the transition zone at Levack there is a concentration of titaniferous magnetite.

Comparison of the results from Azilda, Blezard, and Creighton (Fig. 3) shows that κ is higher on the extreme southern edge of the norite, in each case, than on the northern edge. This would, of course, be consistent with some differentiation of the magnetite across the norite. The rise in the value of κ toward the center could be attributed to either an increase in the percentage composition or in the grain size of the ferromagnetic constituents.

Examination of the Schmidt projections shows that the directions of the RM vectors of the south range specimens are well grouped (see also Fig. 4) and do not differ by more than 40° from the present direction of the earth's field. As a result, the component of the mean RM vector in the direction of the earth's magnetic field will be considerable. The magnetic anomalies observed over the norite section of the nickel irruptive on the south range are therefore due largely to the presence of RM in the rocks, and to a lesser extent to induced magnetization because the value of Q remains well above unity across the three traverses. This statement also appears to be true for the norite on the north

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

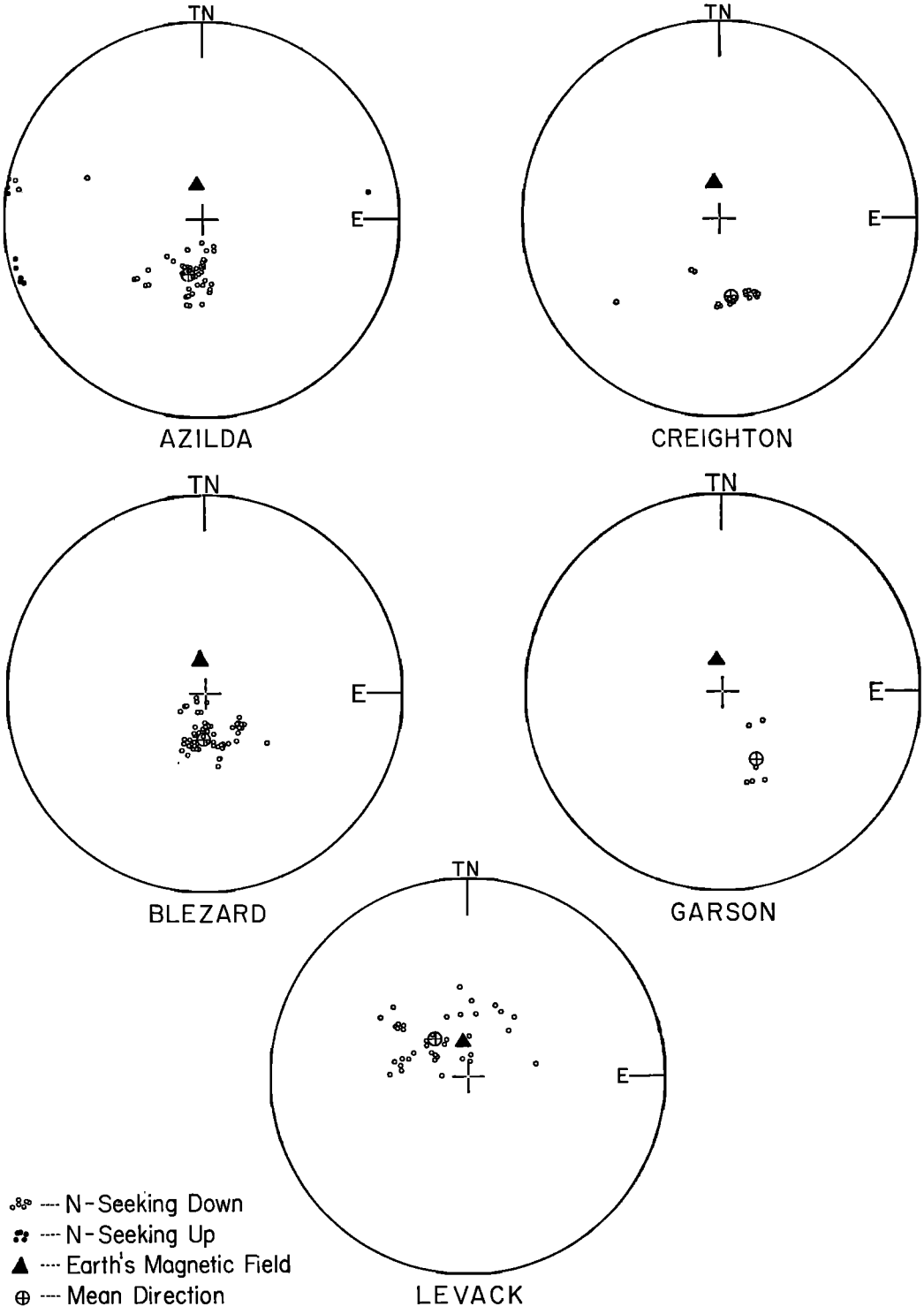


Fig. 2. Schmidt equal-area projections for the five traverses across the nickel irruptive.

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

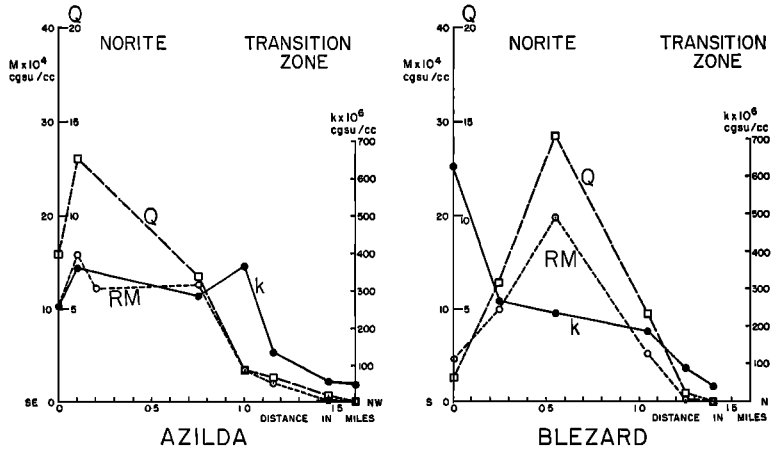
range, i.e., Levack, although this is probably not the case for the transition zone or the micropegmatite, since Q is rather less than unity for these.

Thus these results provide a further example that RM is rather more important than is generally realized by those engaged in the interpretation of aeromagnetic maps. Several articles [*Sutton and Mumme*, 1957; *Hall*, 1959; *Green*, 1960; *Girdler and Peter*, 1960] have recently been published which have also shown that it is often unjustified to assume that the effects of RM can be ignored in the interpretation of aeromagnetic surveys.

Figure 4 is a Schmidt equal-area plot of the combined RM results for the south range. Since the grouping is quite good, and since the mean directions for each of the four traverses fall close to one another, it would appear that there has been very little relative rotary movement across the 16 miles of the nickel irruptive between Creighton and Garson since the irruptive last cooled through the Curie point of its magnetic constituents.

A summary of the paleomagnetic results is given in Table 1, together with the results of the statistical analysis [*Fisher*, 1953] which is usually employed in paleomagnetic investigations.

The Precambrian geomagnetic pole positions have been calculated from the mean directions of the results from Levack and the south range [*Irving*, 1956], and these are also listed in Table 1.



SAMPLE	M x 10⁴ cgsu/cc	K x 10⁶ cgsu/cc	Q = M/KH	ROCK TYPE
18	5 28	298	3·00	NORITE
23	5 86	5080	0·20	TZ
24	1 50	5058	0 05	TZ
26	1 70	51	5·70	NORITE
48	0 82	659	0 21	MP
50	4·40	1124	0 67	MP
53	2 28	3006	0 13	TZ

TZ = TRANSITION ZONE
MP = MICROPEGMATITE

LEVACK

Fig. 3. Susceptibility (κ), remanent magnetization (RM), and the Koenigsberger ratio (Q) for four traverses.

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

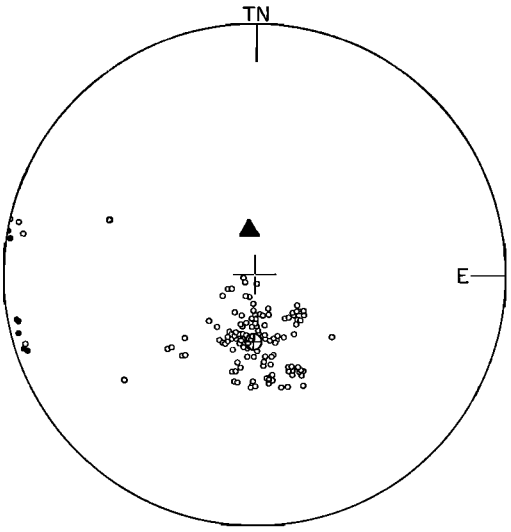


Fig. 4. Schmidt projection for the south range.

The most important result of this investigation is that the mean directions of the samples collected from the north and south ranges differ by some 40°; the mean direction for the north range (Levack) dips to the north, and that for the south range dips to the south.

The geologic interpretation of this result is equivocal. It seems reasonable to assume that the shearing on the south range has not modified the RM directions obtained from the four traverses which extend over a distance of 16 miles. Therefore, if the irruptives on either side of the basin are the same geologic age, it would appear that there has been some relative movement about an east-west axis through the center of the basin, between the irruptive on the north and the irruptive on the south range. In fact, there appears to have been about 40° of relative tilting.

In calculating a pole position for the combined Sudbury results, if it is assumed that there has been equal tilting, i.e. 20°, on either side of the axis, the result labeled (*a*) in Table 1 is obtained. Alternatively, there is some indication from geologic evidence [*Thomson*, 1956] that the south range may have been more affected by tectonic action than the north range, since the latter acted as a foreland. If we therefore assume that the north range was only tilted by 10° and the south range by about 30°, the result (*b*) in Table 1 is obtained. This pole position, however, falls farther away from those obtained by other workers [e.g. *Graham*, 1953; *Irving*, 1957] than does result (*a*).

All these conclusions are based on the assump-

TABLE 1. Paleomagnetic Results from Sudbury

Sample Location	Magnetic Direction						Sampling Area		Pole position			
	Decl.	Incl.	N*	R†	α_{95}‡	K§	Long., °W	Lat., °N	Long., °W	Lat., °N	Oval of Confidence‖ Δp	Δm
Levack	320.4	69.9	41	38.980	5.2	19.8	81.4	46.6	140.7	64.0	8	9
Creighton	171.4	57.0	26	25.464	4.2	46.7	81.2	46.5	...	...	...	...
Azilda	194.4	66.7	45	45.342	2.6	66.8	81.1	46.5	...	...	...	...
Blezard	183.9	71.7	78	77.105	1.7	86.0	81.0	46.6	...	...	...	...
Garson	155.7	59.2	6	5.887	10.2	44.2	80.9	46.6	...	...	...	...
South range combined	182.8	68.0	155	151.865	1.6	49.1	81.0	46.5	82.8	7.6	2	3
North and south ranges combined												
(*a*)	245.3	82.3	196	...	...	...	81.2	46.6	99.4	38.4	...	...
(*b*)	300	78	196	...	...	...	81.2	46.6	115.9	53.4	...	...

* Number of specimen cubes measured.
† Resultant vector of N unit vectors.
‡ Angular radius of the circle of confidence for a probability of 95 per cent.
§ Dispersion.
‖ Δp, semi-major axis; Δm, semi-minor axis.
(*a*) Assuming equal tilting of north and south ranges.
(*b*) Assuming 10° tilt of north range and 30° tilt of south range.

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

tion that the direction of RM observed actually coincided with that of the earth's field at the time the rocks cooled through their Curie points. It is not inconceivable that some other process, as yet undetermined, has produced this difference in the direction of the RM vectors in the north and south ranges. For this and other reasons, it is felt that this paleomagnetic study of the nickel irruptive at Sudbury has really been only a reconnaissance study.

For this study samples from rock outcrops only were collected. Additional information on the tectonic history of the basin might be forthcoming if the intrusive were also sampled at depth and at points closer to the more inaccessible southwest and northeast ends of the basin, and if more mineralogical investigation of the samples were undertaken.

I should like to thank Professors J. T. Wilson and R. M. Farquhar for much encouragement and helpful discussion, the Shell Oil Company Limited for a fellowship during the year 1956–1957, and Hunting Survey Corporation Limited for the loan of a susceptibility meter.

References

Collins, W. H., Life-history of the Sudbury nickel irruptive, 1, Petrogenesis, *Trans. Roy. Soc. Can.*, IV, *28*, 123–178, 1934.

Collins, W. H. and E. D. Kindle, The life history of the Sudbury nickel irruptive II, *Trans. Roy. Soc. Can.*, IV, *29*, 27–38, 1935.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc., London, A, 217*, 295–305, 1953.

Girdler, R. W., and G. Peter, An example of the importance of natural remanent magnetization in the interpretation of magnetic anomalies, *Geophys. Prospecting, 8*, 474–483, 1960.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Research, 54*, 131–167, 1949.

Graham, J. W., Changes of ferromagnetic minerals and their bearings on magnetic properties of rocks. *J. Geophys. Research, 58*, 243–260, 1953.

Green, R., Remanent magnetization and the interpretation of magnetic anomalies, *Geophys. Prospecting, 8*, 98–110, 1960.

Hall, D. H., Direction of polarization determined from magnetic anomalies, *J. Geophys. Research, 64*, 1945–1959, 1959.

Hood, P. J., Palaeomagnetic studies of some Precambrian rocks in Ontario, Unpublished Ph.D. thesis, University of Toronto, 1958.

Irving, E., Palaeomagnetic and palaeoclimatological aspects of polar wandering, *Geofis. pura e appl., 33*, 23–41, 1956.

Irving, E., The origin of the palaeomagnetism of the Torridonian sandstones of N.W. Scotland, *Phil. Trans. Roy. Soc. London, A, 250*, 100–110, 1957.

Lochhead, D. R., The Falconbridge ore deposit, Canada, *Econ. Geol., 50*, 42–50, 1955.

Miller, A. H., and M. J. S. Innes, Gravity in the Sudbury basin and vicinity, *Publs. Dominion Observatory, 18*, 13–43, 1955.

Mooney, H. M., and R. Bleifuss, Magnetic susceptibility measurements in Minnesota, 2, Analysis of field results, *Geophysics, 18*, 383–393, 1953.

Phemister, T. C., Igneous rocks of Sudbury and their relation to the ore deposits, *Ann. Rept. Ontario Dept. Mines, 34*, 1–57, 1925.

Runcorn, S. K., Magnetization of rocks, in *Handbuch der Physik*, Springer-Verlag, Berlin, *47*, 470–497, 1956.

Sutton, D. J., and W. G. Mumme, The effect of remanent magnetization on aeromagnetic interpretation, *Australian J. Phys., 10*, 547–557, 1957.

Thellier, E., and F. Rimbert, Sur l'analyse d'aimantations fossiles par action de champs magnetiques alternatifs, *Compt. rend., 239*, 1399–1401, 1954.

Thomson, J. E., Geology of the Sudbury basin, *Ann. Rept. Ontario Dept. Mines, 65*, 1–56, 1956.

Wetherill, G. W., G. L. Davis, and L. T. Aldrich, Age measurements on rocks north of Lake Huron, *Trans. Am. Geophys. Union, 38*, 412, 1957.

Zietz, I., and R. G. Henderson, The Sudbury aeromagnetic map as a test of interpretation methods, *Geophysics, 20*, 307–317, 1955.

(Manuscript received October 28, 1960;
revised January 10, 1961.)

21562202, 1961, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i004p01235 by Curtin University Library, Wiley Online Library on [29/04/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License